Logarithmic Functions

Exponential function $b > 0$, $b \neq 1$

$$y = b^x$$

Ex: $b=2$ $y = 2^x$ $\log_2(\,)$

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

$\log_2(x)$ is the power y such that $2^y = x$.

$2^3 = 8$

$\log_2(8) = 3$

$\log_2(8)$ is the answer to the question, "What power do I have to raise 2 to, to get 8?"

$y = \log_2(x)$

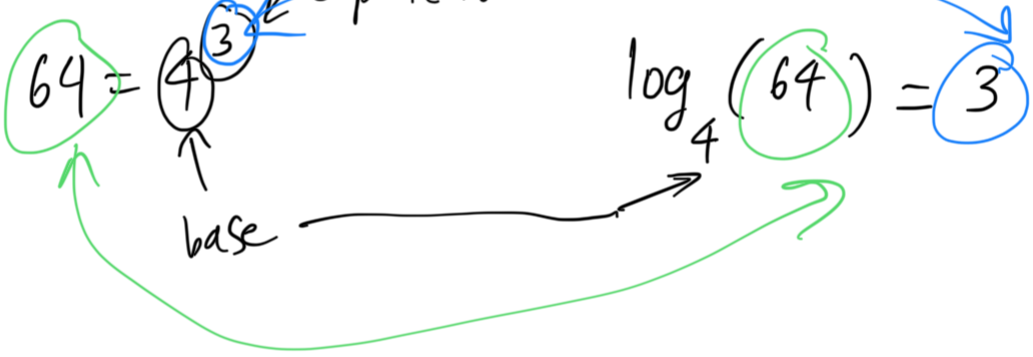
x	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8
y	-3	-2	-1	0	1	2	3

$\log_2\left(\frac{1}{8}\right) = -3$ because $2^{-3} = \frac{1}{8}$

SAME

Exponential

Logarithms



$9 = 3^2$ $\qquad$ $\log_3(9) = 2$

3^x

$x = 2 \rightsquigarrow y = 9$

$\log_3(9) = 2$

$x = 9 \rightsquigarrow y = 2$

order is switched

$10^{-2} = 0.01$ $\qquad$ $\log_{10}(0.01) = -2$

$4^3 = 64$ $\qquad$ $\log_4(64) = 3$

$\log(\ ) = \log_{10}(\ )$

$\log 100 = \log_{10}(100) = 2$ $\qquad$ because $10^2 = 100$

Ex: $\log_2(4) = 2$ $\qquad$ because $2^2 = 4$

Ex: $\log_{10} 10000 = 4$ because $10^4 = 10000$

Ex: $\log_5 1 = 0$ because $5^0 = 1$

If $b > 0$, $b \neq 1$

$\log_b 1 = 0$ because $b^0 = 1$

Ex: $\log_{1257} 1257 = x$ $\quad 1257^1 = 1257^x$

base

$\log_b b = 1$ because $b^1 = b$

$\log_b (b^2) = 2$ because b raised to the power of 2 is b^2.

In general, $\log_b (b^c) = c$

Ex: $\log_{64} \left(\frac{1}{8}\right) = x$ $\quad 64^x = \frac{1}{8}$

$x = -\frac{1}{2}$ $\quad 64^{\frac{1}{2}} = 8$

$64^{-\frac{1}{2}} = \frac{1}{8}$

Ex: $\log_{25} 5 = \frac{1}{2}$ because $25^{\frac{1}{2}} = 5$

Ex: Solve for x

$$\log_x 49 = 2$$

$$x^2 = 49$$

$$x = 7, -7$$

$\log_7 49 = 2$

~~$\log_{(7)} 49 = 2$~~

needs to be > 0.

Since the base has to be positive,

$$x = 7$$

Ex: $\log_8 4 = x$

$$8^x = 4$$

$$(2^3)^x = 2^2$$

$$2^{3x} = 2^2$$

$3x = 2$

$\boxed{x = \frac{2}{3}}$

Ex: $\log_{1000} 10000000 = x$

$1000^x = 10000000$

$\checkmark$ powers of 10

$1000 = 10^3$

$10000000 = 10^7$

$(10^3)^x = 10^7$

$10^{3x} = 10^7$

$3x = 7$

$\boxed{x = \frac{7}{3}}$

Properties of Logarithms

$b > 0, b \neq 1$

$f(x) = b^x$ $\quad$ $g(x) = \log_b(x)$

$f(x)$ and $g(x)$ are inverses of each other.

meaning: $f(g(x)) = x$ (when $x > 0$) } for all x in the domain
and $g(f(x)) = x$ } (all real numbers)

$f(g(x)) = x \rightsquigarrow f(\log_b(x)) = x$

$$b^{\log_b(x)} = x$$

for all $x > 0$.

$g(f(x)) = x \rightsquigarrow g(b^x) = x$

$$\log_b(b^x) = x$$ for all real numbers x.

Ex: $\log_{1254739}\left(1254739^{6512}\right) = 6512$

don't need to know

Ex: $\left(\log_{12579} 86\right)$

$12579^{\cdot} = 86$

Ex: $\log_8 8^2 = 2$

Ex: $3^{\log_3 2} = 2$ $\qquad b^{\log_b x} = x$

Ex: $3^{4\log_3 8} = \left(3^{\log_3 2}\right)^4 = 2^4 = 16$

$a^{mn} = (a^m)^n$

$= (a^n)^m$

$(3^4)^{\log_3 2}$ correct but not helpful

$a^{c\log_a b} = \left(a^{\log_a b}\right)^c = b^c$

Ex: $\log_{10} 0.001^3 = \log_{10}\left[(10^{-3})^3\right]$

10^{-3} rewrite $= \log_{10}(10^{-9})$

0,001 = 10

$\frac{1}{1000}$

as 10 raised to a power $= -9$

CCC 1.0

Ex: $\log_{\frac{1}{2}}(\log_2(16)) = \log_{\frac{1}{2}}(4) = x$

$= -2$

$\left(\frac{1}{2}\right)^x = 4$

$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\left(\frac{1}{4}\right)^{-1} = 4$

$\left(\frac{1}{2}\right)^{-2} = 4$

Ex: $\log_3(9^8) = x$

$3^x = 9^8$

$3^x = (3^2)^8$

$x = 2 \cdot 8$

$3^x = 5$

$x = 2.8$

$\boxed{x = 16}$

Ex: What is $z = \log_3(a^b)$ in terms of $\log_3 a$?

$w = \log_3 a$

$z = \log_3(a^b)$ means that $3^z = a^b$

$w = \log_3 a$ means that $3^w = a$

$3^w = a$ so $(3^w)^b = a^b$

$3^{wb} = a^b = 3^z$

$wb = z$

So $\boxed{b(\log_3 a) = \log_3(a^b)}$